10.1.2 BINOMIAL DISTRIBUTION

Introduction

Binomial distribution was discovered by James Bernoulli (1654 – 1705) in the year 1700 and was first published in 1713 eight years after his death. The distribution of the Sum of n independent Bernoulli variables is known as a Binomial distribution.

That is the sum of outcome of n independent experiments of Bernoulli trials, in each of which the probability of success is constant p and the probability of failure is q=1– p is called Binomial experiment.

Definition

A random variable X denoting the number of successes in an outcome of a Binomial experiment having n trials and p as the probability of success in each trial is called Binomial random variable. Its probability mass function is given by

$$P(X = x) = \begin{cases} nC_x p^x q^{n-x} & for\ x = 0,1,2,...n \\ 0 & otherwise \end{cases}$$ where q=1-p



NOTE

(i) If X is a Binomial variate, it is denoted by $X \sim B(n,p)$

(ii) n,p are the parameters of the distribution.

(iii) The probability values are the successive terms of the expansion of $(q+p)^n$

Conditions for Binomial Distribution

We get the Binomial Distribution under the following experimental conditions:

(i) The number of trials `n' is finite

(ii) The trials are independent of each other

(iii) The probability of success `p' is same for each trial

(iv) Each trial must result in a success or a failure.



NOTE

The problems relating to tossing of coins or throwing of dice or drawing cards with replacement from a pack of cards lead to Binomial distribution.

Characteristics of Binomial Distribution

(i) Binomial distribution is a discrete distribution i.e., X can take values 0, 1, 2, ... n where `n' is finite